A-A-54011
6 November 1989

COMMERCIAL ITEM DESCRIPTION

PLIERS, DENTAL (Amalgam Reinforcement Pin)

The General Services Administration has authorized the use of this Commerical Item Description as a replacement for Military Specification MIL-P-36855A which is cancelled.

This commercial item description covers wire cutting pliers with a plastic insert in the jaws for use in amalgam pin reinforcement technique.

Salient characteristics:

Materials:

Pliers. Pliers shall be fabricated of chrome-vanadium steel AISI 6150 as specified in SAE J404 for composition only, or carbon steel SAE 1055 as specified in SAE J403 for composition only.

Plastic inserts for jaws. The insert shall be resilient, vinyl plastisol.

Plastic handle covering. The handle covering shall be a colored polyvinyl chloride type of plastic.

Hardness.

Steel. After proper stress relieving, the Rockwell hardness shall be C36 to C50 for the body, and C55 to C65 for the cutting edge of the jaws when tested in accordance with ASTM Designation E18 (Rockwell) or ASTM Designation E92 (Vickers) and E140.

Plastisol. Vinyl plastisol (end item, with insert applied to pliers) shall have a hardness equal to a 10 second reading of 50.0 to 75.0 points when tested as specified herein.

Plastisol after aging. After aging, the vinyl plastisol hardness (end item, with insert applied to pliers) shall increase no more than 10.0 points when tested as specified herein.

AMSC N/A FSC 6520

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Distortion. The handles of the pliers shall show no pliability or distortion under normal use.

Style, design and dimensions. The pliers shall be a diagonal cutting type. Pliers shall be specifically designed to be suitable for use in the dental operatory during the treatment of patients, in support of the Markley pin-retention technique. Pliers shall be in accordance with Defense Personnel Support Center (DPSC) Drawing No. 24117. The pliers shall have a plastic insert in each jaw which shall serve as a cushion. The pliers shall be capable of cutting, gripping and holding one cut end of 0.022 and 0.025 inch diameter stainless steel wire.

Construction.

Jaws. The inner jaws of the pliers shall be serrated and the insert shall be primed for good adhesion to steel and affixed to each inner jaw.

Locks. The joint of the pliers shall be the lap joint type. Locks shall be accurately fitted, without stiffness, and without crevices, burrs, or sharp edges (outer edges of joints shall be properly beveled). The remaining edges of the joints shall be properly beveled to prevent pinching of tissue. Pins shall be permanently secured.

Handles. The handles shall be tightly encased in a neat, resilient, smooth-surfaced, plastic which will provide a comfortable, non-skid grip.

Cutting edges. Cutting edges shall be properly sharpened to the required type of edge and shall be free from burrs or feathered edges. The pliers shall function correctly for the particular work for which they are intended. Pliers shall be capable of cutting 0.022 and 0.025 inch diameter stainless steel wire without causing an increase in the diameter of the wire at the cut ends when tested as specified herein.

Finish. Finish on all edges and surfaces shall be uniform and free of burrs, sharp edges (except where required), crevices, grind marks, rough areas, cracks, pits, voids and overlaps. Exposed portions of handles shall be finely ground and polished. The diagonal cutting edges shall be finely honed. The beaks and head shall be polished to a bright finish. As an alternate, the pliers shall be bright chrome plated.

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Identification marking. Marking shall be on the (outside, front or back face) metal portion (joint area). The marking shall be permanent and legible (chemical or electrochemical etching shall be considered acceptable). Pliers made in the U.S.A. shall be marked with the name or registered trademark of the contractor. Pliers of foreign manufacture shall be marked with the name of the country of origin, together with the name or registered trademark of the contractor.

Workmanship. The pliers shall be free from defects which detract from their appearance or impair their serviceability, proper functioning and intended use.

Unit. Each (EA). One pliers, as specified, constitutes one unit.

Unit package. Each pair of pliers shall be enclosed in a 4x7 inch plastic bag. The plastic bag shall then be enclosed in a 3-1/2 x 6-1/2 inch paper envelope. The flap shall be sealed.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the used of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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Certificates of quality. Certificates of quality for raw material, supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government.

Sampling for inspection. Sampling for inspection shall be conducted in accordance with MIL-STD-105. Where feasible, the same sample shall be used for the determination of two or more inspection, examination and test characteristics. The unit of product for inspection, examination and test purposes shall be one pliers. Unless otherwise specified, sampling for examination shall be conducted at inspection level II, except for dimensions which shall be conducted at inspection level S-1, and sampling for tests (when applicable shall be conducted at inspection level S-2).

Examination. Examination shall include, but shall not be limited to the following:

Defects

Style and design not as specified.
Jaw plastic insert inadequately secured, loose or missing.
Handles not tightly encased in plastic.
Handle plastic not resilient and smooth surfaced.
Metal surface finish not as specified.
Edges and surfaces not free from burrs, crevices, pits, voids and overlaps.
Plastic parts not free from patches, cracks, holes, voids, or other imperfections.
Joint of pliers not free from stiffness.
Joint of pliers has excessive play or is excessively loose.
Edges of the joint of pliers sharp (not properly beveled).
Cutting edges not sharp.
Cutting edges not properly aligned.
Members of pliers not permanently and legibly marked with the name or trademark of the contractor.
Marking not properly located.
Metal surfaces not free from grind marks and rough areas.
Plating not free of peeling or blisters.
Dimensions not as specified.

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Tests.

Chemical analysis. In the absence of certificates of quality, chemical composition of the steel used in the manufacture of the pliers shall be determined in accordance with method 111 or 112 of FED-STD-151. In the event of a conflict, method 111 shall govern.

Plastisol tests. The following plastisol tests shall be performed after the plastisol inserts or cushions have been applied to the throat of the jaws:

Hardness test. The hardness characteristic of the vinyl plastisol shall be tested using a Shore Type A-2 Durometer, or equivalent testing equipment. Firm pressure shall be used to seat the instrument against the specimen. Penetration shall be maintained and the reading noted at 10 seconds. Three readings shall be averaged for each unit of product tested. Defective readings shall be discarded and additional readings taken. A new spot shall be penetrated for each reading.

Simulated aging test. After being oven aged at 158.0 $\pm$ 4.0 degrees (^{0}F) for 48 hours, the sample shall be removed from the oven and allowed to cool to room temperature. The hardness test shall again be performed.

Cutting test. The pliers shall cut 0.022 and 0.025 inch diameter stainless steel wire as specified. Three separate consecutive tests shall be performed for each pliers. Each test shall consist of making one cut through the test material in a different portion of the jaws.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producer's own drawings, specifications, standards and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch/pounds units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and of regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document.

National Stock Number	Item Identification
6520-00-935-1094	PLIERS, DENTAL, Amalgam Reinforcement Pin, 4-7/8 inch: With plastic insert in each jaw.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be as specified in the contract and/or order.

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MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

CIVIL AGENCY COORDINATING ACTIVITIES:
VA-OSS
PHS
FDA-MPQAS

48636

PREPARING ACTIVITY:

DoD-MB

AGENT:

DLA-DM

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